ADJUSTING PRODUCT REPRESENTATIONS FOR SEQUENTIAL RECOMMENDER SYSTEMS

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June 27th 2025

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INTRODUCTION

Sequential recommender systems predict a user's next interaction based on past behavior. ID-based models like SASRec perform well but struggle with cold-start and generalization.
Recformer replaces item IDs with product text, using a language modeling approach for richer semantic understanding.
We reproduce Recformer and compare it with ID-only, hybrid, and text-only baselines on Amazon and MIND. We evaluate reproducibility, generalization, and diversity, and extend Recformer with sentiment features from user reviews (Recformer-ABSA).

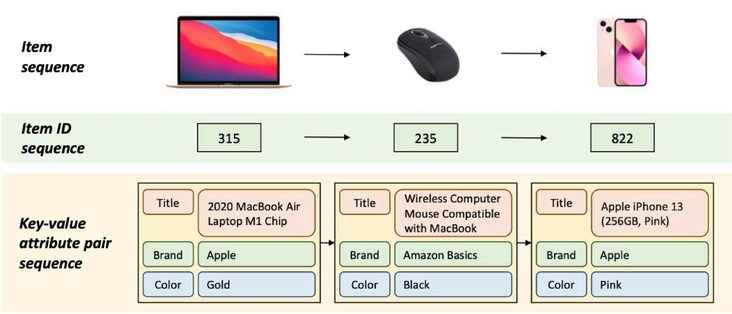
BACKGROUND & RELATED WORK

Sequential recommender systems predict future user behavior by modeling the order of past interactions, offering improvements over static recommendation methods.
Recent models improve performance by using separate attention layers for item features and by introducing soft labels instead of one-hot encoding.
In addition to accuracy, recommendation diversity is increasingly important. Metrics like intra-list diversity and category coverage help ensure more varied and user-friendly recommendations.

METHOD

We evaluate four models, each representing a different use of item information:

- SASRec (ID-only):
  Uses self-attention over item ID sequences without any metadata.
- FDSA (Hybrid):
  Combines item IDs with side features (e.g., brand, category) using separate attention layers.
- UniSRec (Text-only):
  Generates item embeddings from product text using a frozen language model and a lightweight sequential layer.
- Recformer (Text-only, end-to-end):
  Treats recommendation as a language modeling task, using item descriptions and metadata in a Longformer-based architecture.



SETUP

We use two datasets: **Amazon Reviews** and **MIND**. Models are pre-trained on multiple categories and fine-tuned on two per dataset.
Performance is evaluated using accuracy (NDCG, Recall, MRR) and diversity (Gini, ILD, CC) metrics. Training follows the original Recformer setup with minor adjustments due to hardware limits.

	Amazon Review Dataset	MIND Dataset
Number of users	3,501,528	477,029
Number of items	1,022,275	34,684
Number of interactions	32,291,280	1,344,391
Average actions per user	9.22	2.82
Average actions per item	33.82	209.21
Sparsity (%)	99.9991%	99.9919%

REPRODUCTION

Our reproduced results on the Amazon dataset match the original performance trends, though absolute scores are slightly lower due to computational constraints.
On the MIND dataset, overall performance improved due to its smaller size, but Recformer underperformed compared to other models, suggesting limited generalization to new domains.

Dataset	Metric	SASRec	FDSA[3]	UniSRec	Recformer	Difference Recformer
Scientific	NDCG@10	0.0685	0.0545	0.0674	**0.1042**	+0.0015
	Recall@10	0.0963	0.0820	0.1269	**0.1465**	+0.0017
	MRR	0.0595	0.0458	0.0490	**0.0968**	+0.0017
Instruments	NDCG@10	0.0972	0.0761	0.0703	**0.0841**	+0.0011
	Recall@10	0.1235	0.1003	0.1249	**0.1054**	+0.0002
	MRR	0.0891	0.0687	0.0533	**0.0821**	+0.0014

Dataset	Metric	SASRec	FDSA	UniSRec	Recformer
MIND + TV	NDCG@10	0.3247	0.3414	**0.3822**	0.3124
	Recall@10	0.5724	0.5900	**0.6242**	0.4609
	MRR	0.2527	0.2656	0.2666	**0.2770**
MIND + Music	NDCG@10	0.3515	0.3610	**0.3931**	0.2016
	Recall@10	**0.6546**	0.6373	0.5926	0.3420
	MRR	0.2594	0.2758	**0.3224**	0.1682

EXTENSION: ABSA

We extended Recformer by adding aspect-based sentiment from Amazon reviews, using a BERT-based sentiment model. Sentiment-labeled aspects (e.g., "design: Positive") were appended to product titles.
Despite the richer input, this enhancement slightly reduced performance, suggesting that sentiment signals from reviews may not improve recommendation quality in this setup.

ABSA FEATURE

```
"4841932909": {
"title": "1013A/Organic Chemistry Set for Student
Edition 2 |
design: Positive, exercises: Positive, material:
Negative,
plastic bag: Negative, packaging: Negative",
"brand": "HGS Maruzen",
"category": "Industrial & Scientific Science
Education"
}
```

DIVERSITY & FAIRNESS

We assessed model diversity using Gini, ILD, and Catalog Coverage. On Amazon, Recformer showed more balanced item distribution (lower Gini) than UniSRec. On MIND, Recformer again had the lowest Gini but also the lowest ILD, indicating less diversity within individual recommendation lists.
UniSRec showed the highest ILD but less fairness overall. SASRec and FDSA offered more balanced trade-offs. These results highlight the need to optimize for both accuracy and diversity in recommender systems.

Model	Input	Dataset	Gini@10	ILD@10	CC@10
SASRec	Id only	MIND + tv	0.6634	0.6215	0.7000
FDSA	Id + text	MIND + tv	0.6277	0.5752	0.6000
UniSRec	text only	MIND + tv	0.6886	0.6615	0.7000
Recformer	text only	MIND + tv	0.3565	0.4902	0.3000
SASRec	Id only	MIND + music	0.6140	0.6259	0.5454
FDSA	Id + text	MIND + music	0.5892	0.6393	0.5454
UniSRec	text only	MIND + music	0.6318	0.6871	0.5000
Recformer	text only	MIND + music	0.3830	0.3385	0.3636

Model	Input	Gini@10
UniSRec	text only	0.7893
Recformer	text only	0.7460

CROSS-DOMAIN INFERENCE

We tested whether models pretrained on Amazon could generalize to the MIND news dataset. All models showed a significant drop in performance, indicating limited cross-domain transferability.

Model	NDCG@10	Recall@10	MRR@10	Gini@10
SASRec	0.0189	0.0351	0.0141	0.752
FDSA	0.0071	0.0137	0.0051	0.7868
UniSRec	0.0221	0.00294	0.02312	0.7823
Recformer	0.026	0.0312	0.0259	0.7733

CONCLUSION

We reproduced Recformer and three baselines (SASRec, FDSA, UniSRec) on Amazon and MIND to assess reproducibility, generalization, and diversity.
Recformer performs best in-domain, but its advantage does not extend consistently to other domains.
Diversity analysis shows Recformer offers more balanced recommendations globally but lower intra-list diversity. UniSRec exhibits the opposite trend.
We also extended Recformer with aspect-based sentiment from reviews. This enriched input did not improve performance, indicating that sentiment data may add noise without more targeted modeling.
These findings highlight the strengths of text-based models, while emphasizing challenges in generalization, diversity, and input enrichment.